CS4001NT Programming

Islington College,Nepal

Document Details

Submission ID

trn:oid:::3618:96176111

Submission Date

May 16, 2025, 10:36 AM GMT+5:45

Download Date

May 16, 2025, 10:41 AM GMT+5:45

File Name

CS4001NT Programming

File Size

45.1 KB

59 Pages

6,513 Words

38,528 Characters

12% Overall Similarity

The combined total of all matches, including overlapping sources, for each database.

Match Groups

- **65** Not Cited or Quoted 10%
  Matches with neither in-text citation nor quotation marks
- **3** Missing Quotations 0%
  Matches that are still very similar to source material
- **11** Missing Citation 1%
  Matches that have quotation marks, but no in-text citation
- **0** Cited and Quoted 0%
  Matches with in-text citation present, but no quotation marks

Top Sources

0%	Internet sources
0%	Publications
12%	Submitted works (Student Papers)

Integrity Flags

0 Integrity Flags for Review

Our system's algorithms look deeply at a document for any inconsistencies that would set it apart from a normal submission. If we notice something strange, we flag it for you to review.

A Flag is not necessarily an indicator of a problem. However, we'd recommend you focus your attention there for further review.

Match Groups

- **65** Not Cited or Quoted 10%
  Matches with neither in-text citation nor quotation marks
- **3** Missing Quotations 0%
  Matches that are still very similar to source material
- **11** Missing Citation 1%
  Matches that have quotation marks, but no in-text citation
- **0** Cited and Quoted 0%
  Matches with in-text citation present, but no quotation marks

Top Sources

0% Internet sources
0% Publications
12% Submitted works (Student Papers)

Top Sources

The sources with the highest number of matches within the submission. Overlapping sources will not be displayed.

1	Submitted works	**islingtoncollege on 2025-05-16**	**4%**
2	Submitted works	**iacademy on 2025-05-14**	**2%**
3	Submitted works	**Asia Pacific University College of Technology and Innovation (UCTI) on 2022-06-19**	**1%**
4	Submitted works	**Southern New Hampshire University - Continuing Education on 2020-04-26**	**<1%**
5	Submitted works	**AlHussein Technical University on 2024-01-31**	**<1%**
6	Submitted works	**The College of West Anglia on 2024-04-18**	**<1%**
7	Submitted works	**Chester College of Higher Education on 2015-05-07**	**<1%**
8	Submitted works	**University of North Texas on 2023-03-24**	**<1%**
9	Submitted works	**Asia Pacific University College of Technology and Innovation (UCTI) on 2018-02-26**	**<1%**
10	Submitted works	**Milton Keynes College on 2019-05-07**	**<1%**

11 Submitted works

TMC Education Group on 2012-03-17 <1%

12 Submitted works

University of Greenwich on 2013-01-09 <1%

13 Internet

www.citethisforme.com <1%

14 Submitted works

Asia e University on 2024-01-25 <1%

15 Internet

bloggingrepublic.com <1%

16 Submitted works

HTM (Haridus- ja Teadusministeerium) on 2023-12-25 <1%

17 Submitted works

Illinois Mathematics and Science Academy on 2023-12-15 <1%

18 Submitted works

Rivier University on 2023-12-09 <1%

19 Submitted works

United Colleges Group on 2025-04-30 <1%

20 Submitted works

University of Sunderland on 2025-03-11 <1%

21 Submitted works

Bangkok Patana School on 2016-02-29 <1%

Introduction

Java is a popular programming language. Many companies use Java to develop app, desktop app, games and many more. For this gym management system I have also used java to perform this task. This project was given as the coursework of 1st year student of programming. The gym Member consists of Gym Member which is the super class followed by Regular member and premium Member where gym member is the abstract class which have a abstract method name mark attendance which helps the program to keep the child class attendance differently according to their needs. The application includes a graphical user interface built with Java Swing.

Figure 1: Java

Tools used

15 Blue J is a free and open-source Integrated Development Environment (IDE) for Java.
1 It was designed for beginners who want to learn java. It was developed by Michael Kolling and John Rosenberg at Monash University, Australia. This project started in late 1990s as a port of “Blue” pedagogical language created by Michael Kolling as part of his PhD work. The first version of Blue J 1.0 was released on August 23,1999. The project was support from Microsystems later Oracle acquired it. The latest version of Blue J is 5.4.1 which was released in Sep 19,2024.

Figure 2:BLUEJ

Aims and Objective

This report represents the gym management system given as coursework for the second semester student. The aim of the coursework is to implement the real-word scenario which consist of parent class Gym Member followed by Regular Member and Premium Member which have unique attributes and common attributes as followed by Gym Member.

The Objectives of the project are as follows: -

To learn and implementation of GUI using swing and awt.

To learn the concepts and implement in real-life scenario.

To learn about button action Listener and to implement exception in real-life scenario.

To learn to making design and layout.

GUI wireframe

Figure 3:Gym GUI

12 Class Diagram

Class Diagram is a type of static structure that describes the structure of a system classes, their attributes, and method. In this project the class diagram represents the structure of Gym Management System and the relationship between their classes. This project is totally based on Object Oriented principal such as inheritance, encapsulation, abstraction, etc.

Gym Member class Diagram

Table 1Gym Member class Diagram:

Regular Member

Table 2:Regular Member class Diagram

Premium Member

Table 3:Premium Member class Diagram

Gym GUI

Table 4: GymGUI class diagram

Final class Diagram

(Sovereign Corporate Tower, 2025)

Table 5: Final class Diagram

Pseudocode

Pseudocode is half English and half code of writing a program to express the program logic. For my project I have created whole program to explain and express the program logic.

Gym Member

CREATE a parent class which is also a abstract class GymMember

DECLARE instance variable id as int with protected access

DECLARE instance variable name as String with protected access

DECLARE instance variable location as String with protected access

DECLARE instance variable phone as String with protected access

DECLARE instance variable email as String with protected access

DECLARE instance variable gender as String with protected access

DECLARE instance variable DOB as String with protected access

DECLARE instance variable memberShipStartDate as String with protected access

DECLARE instance variable attendance as int with protected access

DECLARE instance variable loyaltyPoints as double with protected access

DECLARE instance variable activeStatus as Boolean with protected access

1 CONSTRUCTOR GymMember(id, name, location, phone, email, gender, DOB, membershipStartDate, attendance, loyaltyPoints, activeStatus)

SET this.id = id

SET this.name= name

SET this.location = location

SET this.phone= phone

SET this.email= email

SET this.gender= gender

SET this.DOB = DOB

```
SET this.membershipStartDate = membershipStartDate
SET this.attendance = attendance
SET this.loyaltyPoints= loyaltyPoints
SET this.activeStatus= activeStatus
END CONSTRUCTOR
ABSTRACT FUNCTION markAttendance()
END FUNCTION
FUNCTION activateMembership()
     SET activeStatus = true
END FUNCTION
FUNCTION deactiveMembership()
IF activeStatus IS true THEN
    SET activeStatus = false
END FUNCTION
FUNCTION resetMember()
   SET activeStatus = false
   SET attendance = 0
   SET loyaltyPoints = 0
END FUNCTION
FUNCTION display()
            OUTPUT “Member ID” + id
            OUTPUT “Name” + name
OUTPUT “location” + location
```

OUTPUT "phone" + phone

OUTPUT "email" + email

OUTPUT "gender" + gender

OUTPUT "DOB" + DOB

OUTPUT "membershipStartDate" +memberShipStartDate

OUTPUT "attendance" + attendance

OUTPUT "loyaltyPoints" + loyaltyPoints

OUTPUT "activeStatus" + activeStatus

END FUNCTION

END CLASS

Regular Member

CREATE a child class as RegularMember

DECARE instance variable as attendanceLImit as int with private access modifier as it has final static value

DECLARE instance variable isEligibleForUpgrade as Boolean with private access modifier

DECLARE instance variable removalReason as String with private access modifier

DECLARE instance variable referralSource as String with private access modifier

DECLARE instance variable plan as String with private access modifier

DECLARE instance variable price as double with private access modifier

```
CONSTRUCTOR RegularMember(id,name,location,phone,email,gender,DOB,
membershipStartDate,referralSource)
CALL SUPER CONSTRUCTOR (id,name,location,phone,email,gender,DOB,
membershipStartDate)
SET this.attendanceLimit = attendanceLimit
SET this.isEligibleForUpgrade = isELigibleForUpgrade
SET this.removalReason = removalReason
SET this.referralSource = referralSourcee
SET this.plan = plan
SET this.price = price
END CONSTRUCTOR


FUNCTION markAttendance()
INCREMENT attendance by 1
INCREMENT loyaltyPoints by 5
IF attence is greater  or equal then attendanceLimit
   SET isELigibleForUpgrade = true
END FUNCTION

FUNCTION getPlanPrice(Strring Plan)
SWITCH plan to lowercase
Case=basic
```

```
  RETURN 6500
Case =Standard
 RETURN 12500
Case Deluxe
 RETURN 18500
DEFAULT
RETURN -1

FUNCTION upgradePlan (String plan)
IF not isEligibleForUpgrade THEN
RETURN “Not eligible for upgrade. Need more attendance”
END IF
IF new.plan to lowercase() equals plan then
RETURN “Already subscribed to this plan”
SETthis. newprice= getPlanPrice
IF new price  == -1 Then
RETURN “Invalid Plan Selected”
SET this.Plan =  newPlan to lowercase
SET this.Price = new Price
RETURN “Plan upgrade successfully to” + newPlan

FUNCTION revertRegularMember(String removalReason)
CALL SUPER  resetMember() method
```

SET this.isEligibleForUpgrade = false

SET this.plan = basic

SET this.price = 6500

SET this.removalReason = removalReason

END FUNCTION

FUNCTION display()

CALL SUPER display() method

OUTPUT “plan”+ plan

OUTPUT “Price”+ price

IF not removal reason is empty then

OUTPUT “removal reason” + removal reason

Premium Member

CREATE a child class name as Premium Member

DECLARE instance variable premiumCharge as double which has final value = 50000 with private access modifier

4 DECLARE instance variable personalTrainer as String with private access modifier

4 DECLARE instance variable isFullPayement as Boolean with private access modifier

4 DECLARE instance variable paidAmount as double with private access modifier

DECLARE instance variable discountAmount as double with private access modifier

1 CREATE CONSTRUCTOR PremiumMember(id, name, location, phone, email, gender , DOB, memberShipStartDate, personalTrainer)

```
CALLSUPERCONSTRUCTOR(id,name,location,phone,email,gender,DOB,membershi
pStartDate)
SET this.premiumCharge  = premiumCharge
SET this.personalTrainer = personalTrainer
SET this.isFullPayemet = isFullPayement
SET this.paidAmount = paidAmount
SET this.discountAmount = discountAmount
END CONSTRUCTOR


FUNCTION  payDueAmount()
IF this.isFullPayement is true then
RETURN “Payement is already complete”
IF this.paidAmount + amount > premiumCharge then
RETURN “payement amount exceeds the premium charge”
INCREMENT paidAmount by amount
IF paidAmount == premiumCharge THEN
SET isFullPayement = true
SET remainigAmount  = PremiumCharge – paidAmount
RETURN “Payment Successful. Remaining amount”’ + Remaining amount
END FUNCTION


FUNCTION calculateDiscount()
```

```
IF isFullPayment is true THEN
SET discountamount = premiumCharge – paidAmount
OUTPUT “Discount calculated” + discountAmount
END IF
OUTPUT “No discount available. Complete Payment first “
END FUNCTION


FUNCTION revertPremiumMember()
CALL SUPER resetMember() method
SET personalTrainer = “”
SET isFullPayement = false
SET paidAmount =0
SET discountAmount = 0
END FUNCTION
FUNCTION display()
CALL SUPER display() method
DISPLAY “Personal Trainer “ + personalTrainer
DISPLAY “Paid Amount” + paidAmount
DISPLAY “ Payment  Status ” +isFullPayement
SET remainingAmount = premiumCharge – paidAmount
DISPLAY “Remaining amount” + remainingAmount
IF isFullPayment is true THEN
DISPLAY “Discount Amount “ + discountAmount
```

END FUNCTION

END CLASS

Gym GUI

Create a child class as GymGUI

DECLARE frame as JFrame

DECLARE idField, nameField, locationField, phoneField, emailField, referralField, paidField, removalField, trainerField reguPriceField, preCharField, diaAmountField as JTextField

DECLARE dobYearBox, dobMonthBox, dobDayBox as JComboBox<String>

DECLARE msYearBox, msMonthBox, msDayBox as JComboBox<String>

DECLARE genderGroup as ButtonGroup

DECLARE malebutton, femalebutton, otherbutton as JRadioButton

DECLARE members as Arraylist<GymMember>

DECLARE regularmembtn, premiumbtn, activateMembershipBtn, deactivateMembershipBtn, markAttendancebtn, revertRegularMemberBtn, revertPremiumBtn, upgradeButton , discountButton as JButton

FUNCTION GymGUI()

CALL createGUI()

END FUNCTION

FUNCTION createGUI()

CREATE JFrame frame ("Gym Management System")

SET frame defaultCloseOperation to JFrame.EXIT_ON_CLOSE

SET layout of frame to null

SET size of the layout (width 900, height 900)

SET background color to rgba(240,240,240)

SET frame contentPane to the same as backgroundcolor

CREATE JPanel headerPanel

SET layout of headerPanel to null

SET headerPanel bounds to (x=0,y=0,width=900,height=50)

SET headerPanel color to rgb(42,68,80)

ADD headerPanel to frame

CREATE JLabel titleLabel ("Gym Management System")

SET bounds to titleLabel(x=300,y=20,width=900,height=50)

SET foreground color to titleLabel to Color.WHITE

SET titleLabel font and size to ("serif",Font.BOLD,14)

SET a light gray line border around the main panel

ADD titleLabel to headerPanel

CREATE a new JPanel name as mainPanel

SET layout of main panel to null

SET bounds of mainPanel to x:30,y:100,width:820,height:410

SET background color of mainPanel to white

SET border of mainPanel to a line border with color (200,200,200)

ADD mainPanel to the frame

CREATE a new JLabel named meminfo with text “member information”

SET bounds of membinfo to x:10,y:10,width:380,height:25

SET font of membinfo to “Segoe UI”, style bold, size 16

ADD membinfo to mainPanel

CREATE a new JLabel named as idLabel with text “Id”

SET bounds of idLabel to x:20,y:50,width:150,height=25

Add idLabel to mainPanel

CREATE a new JTextField names as idField

SET bounds to idField to x:180,y:50,width:200,height=30

ADD idField to mainPanel

CREATE a new JLabel named as nameLabel with text “name”

SET bounds to nameLabel to x:20,y:90,width:150:height:30

ADD nameField to mainPanel

CREATE a new JTextField named as nameField

SET bounds to nameField to x:180,y:50,width:200,height:30

ADD nameField to mainPanel

CREATE a new JLabel name as locationLabel with text “location”

SET bounds to locationLabel to x:20,y:130,width:150,height:30

ADD locationLabel to mainPanel

CREATE a new JTextField named locationField

SET bounds to locationField to x:180,y:130,width:200,height:30

ADD locationField to mainPanel

CREATE a new JLabel named as phoneLabel with text “Phone”

SET bounds to phoneLabel to x:20,y:170,width:200,height:30

ADD phoneLabel to mainPanel

CREATE a new JTextField named as phoneField

SET bounds to phoneField to x:180, y:170, width:150, height:30

ADD phoneField to mainPanel

CREATE a new JLabel named emailLabel with text “email”

SET bounds to emailLabel to x:20,y:210,width:150,height:30

ADD emailLabel to mainPanel

CREATE a new JTextField named emailField

SET bounds to emailField to x:180,y:210,width:200,height:30

ADD emailField to mainPanel

CREATE a new JLabel named as genderlabel with text “gender”

SET bounds to genderLabel to x:20,y:250,width:150,height:30

ADD genderLabel to mainPanel

CREATE a new JRadioButton name as malebutton with text “male”

SET bounds to malebutton to x:180,y:250,width:100,height:30

SET background color of malebutton to white

ADD male button to mainPanel

CREATE a new JRadioButton named as femalebutton with text “female”

SET bounds to femalebutton to x:250,y:250,width:100,height:30

SET background color of femalebutton to white

ADD femalebutton to mainPanel

CREATE a new JRadiobutton named as otherbutton with text “other”

SET bounds to otherbutton to x:320,y:250,width:100,height:30

SET background color of otherbutton to white

ADD otherButton to mainPanel

17 CREATE a new ButtonGroup named as genderGroup

ADD malebutton to genderGroup

ADD femalebutton to genderGroup

ADD otherbutton to genderGroup

CREATE a new JLabel named as dobLabel with the text “DOB”

SET bounds to dobLabel to x:20,y:250,width:140,height:30

ADD dobLabel to mainPanel

CREATE a JComboBox named as dobYearBox with items from getYears()

SET bounds to dobYearBox to x:20,y:290,width:150,height:30

ADD dobYearBox to mainPanel

CREATE a JComboBox named as dobMonthBox with items from getMonth()

SET bounds to dobMonthBox to x: 250,y:285,width:60,height:25

ADD dobMonthBox to mainPanel

CREATE a JComboBox named as dobDayBox with items from getDay()

SET bounds to dobDayBox to x:315,y:285,width:60,height:25

ADD dobDayBox to mainPanel

CREATE a new JLabel named as membershipLabel with text “Membership Details”

SET bounds to membershipLabel to x:440,y:10,width:380,height:25

SET font of membershipLabel to “segoe UI”,style Bold, size:16

ADD membershipLabel to mainPanel

CREATE a new JLabel named as membershipStart with text “Membership Start”

SET bounds to membershipStart to x:440,y:50,width:150,height:30

ADD membershipStart to mainPanel

CREATE a JComboBox named as msYearBox

SET bounds to msYearBox to x:600,y:50,width:60,height:25

ADD msYearBox to mainPanel

CREATE a JComboBox named as msMonthBox

SET bounds to msMonthBox to x:670,y:50,width:60,height:25

ADD msMonthBox to mainPanel

CREATE a JComboBox named as msDayBox

SET bounds to msDayBox to x:750,y:50,width:60,height:25

ADD msDayBox to mainPanel

CREATE a JLabel named as referralLabel with text “Referral Source”

SET bounds to referralLabel to x:440,y:100,width:160,height:30

ADD referralLabel to mainPanel

CREATE a new JTextField named as referralField

SET bounds to referralField to x:600,y:100,width:200,height:30

ADD referralField to mainPanel

CREATE a new JLabel named as paidLabel with text “Paid Amount”

SET bounds to paidLabel to x :440,y:140,width:150,height:30

ADD paidLabel to mainPanel

CREATE a new JTextField named as paidField

SET bounds to paidField to x;600,y:140,width:200,height:30

ADD paidField to mainLabel

CREATE a new JLabel named as removLabel text as “Removal Reason”

SET bounds to removLabel to x:440,y:180,width:150,height:30

ADD removLabel to mainPanel

CREATE a new JTextField named as removalField

SET bounds to removalField to x:600,y:180,width:200,height:30

ADD removalField to mainPanel

CREATE a new JLabel named as trainerLabel with text “Trainer Name”

SET bounds to trainerLabel to x:440,y:220,width:150,height:30

ADD trainerLabel to mainPanel

CREATE a new JTextField named as trainerField

SET bounds to trainerField to x:600,y:220,width:200,height:30

ADD trainerField to mainPanel

CREATE a new JLabel named as planLabel with text “Membership Plan”

SET bounds to planLabel to x:440,260,150,30

ADD planLabel to mainPanel

CREATE a String array named plan as values “basic” “standard” “deluxe”

CREATE a JComboBox named as planBox

SET bounds to planBox to x:600,y:250,width:200,height:25

SET background color of planBox to white

ADD planBox to mainPanel

CREATE a JLabel named discountLabel with text “Discount Amount”

SET bounds to discountLabel to x:440,y:300,width:150,height:30

ADD discountLabel to mainPanel

CREATE a JTextField named disAmounField

SET bounds to disAmounField to x:600,y:300,width:200,height:30

ADD disAmounField to mainPanel

CREATE a new JLabel named regularPlanPrice with text “Regular Plan Price”

SET bounds to regularPlanPrice to x:440,y:340,width:150,height:30

ADD regularPlanPrice to mainPanel

CREATE a new JTextField named reguPriceField

SET bounds to reguPriceField to x:600,y:340,width:200,height:30

SET text of reguPriceField to “6500”

SET editable changes of reguPriceField be false

ADD reguPriceField to mainPanel

CREATE a new JLabel named premiumPlanPrice with text “Premium Plan Price”

SET bounds to premiumPlanPrice to x:440,y:380,width:150,height:30

ADD premiumPlanPrice to mainPanel

CREATE a new JTextField named preCharField

SET bounds to preCharField to x:600,y:380,width:200,height:30

SET text to preCharField as “50000”

SET editable text to fase

ADD preCharField to mainPanel

CREATE a new JPanel named as buttonPanel

SET layout of buttonPanel to null

SET bounds to buttonPanel to x:30,y:500,width:820,height:400

ADD buttonPanel to frame

CREATE a JButton named as regularmemBtn with text “ Add Regular Member”

6 SET bounds to regularmemBtn to x:30,y:30,width:200,height:30

SET background color to headerColor

SET font of regularmemBtn to “Poppins”, style PLAIN, size 14

ADD regularmemBtn to buttonPanel

CREATE a JButton named as premiumBtn with text “Add Premium Member”

SET bounds to premiumBtn to x:300,y:30,width:200,height:30

SET font of premiumBtn to “Poppins”, style PLAIN, size 14

ADD premiumBtn to buttonPanel

WHEN regularmemBtn is clicked:

CALL addRegularMember() method

WHEN premiumBtn is clicked:

CALL addPremiumMember() method

CREATE a new JButton named activateMembershipBtn with text “Activate Membership”

SET bounds to activateMembershipBtn to x:600,y:30,width:200,height:30

SET font of activateMembershipBtn to “Poppins”, Style.PLAIN, 14

ADD activateMembershipBtn to buttonPanel

WHEN activateMembershipBtn is clicked:

CALL activateMembershipBtn() method

CREATE a new JButton named as deactiveMembershipButton with text “ Deactivate Membership”

6 SET bounds to deactivateMembershipButton to x:30,y:90,width:200,height:30

SET font to deactivateMembershipButton to “Poppins”,Style.PLAIN,14

ADD deactivateMembershipButton to buttonPanel

WHEN deactivateMembershipButton is clicked:

CALL deactivateMembershipButton() method

CREATE a JButton named markAttendancebtn with text “Mark Attendance”

SET bounds to markAttendancebtn to x:300,y:90,width:200,height:30

SET font to markAttendance to “Poppins”,Style.PLAIN,14

ADD markAttendancebtn to buttonPanel

WHEN markAttendancebtn is clicked:

CALL addmarkAttendancebtn() method

CREATE a JButton named revertRegularMemberbtn with text “Revert Regular Member”

SET bounds to revertRegularMemberbtn to x:600,y:90,width:200,height:30

SET font to revertRegularMemberbtn to “Poppins”,Style.PLAIN,14

ADD revertRegularMemberbtn to buttonPanel

WHEN revertRegularMemberbtn is clicked:

CALL addrevertRegularMemberbtn() method

CREATE a JButton named revertPremiumMember with text “Revert Premium Member”

6 SET bounds to revertPremiumMember to x:30,y:150,width:200,height:30

SET font to revertPremiumMemberbtn to “Poppins”,Style.PLAIN,14

ADD revertPremiumMemberbtn to buttonPanel

WHEN revertPremiumMemberbtn is clicked:

CALL addrevertPremiumMemberbtn() method

CREATE a new JButton named displayMemberButton with text “Display Members”

SET bounds to displayMemberButton to x:300,y:150,width:200,height:30

SET font to displayMemberButton to “Poppins”,Style.PLAIN,14

ADD displayMemberButton to buttonPanel

WHEN displayMemberButton is clicked:

CALL displayMembers() method.

CREATE a new JButton named clearFieldsButton with text “Clear Fields”

SET bounds to clearFieldsButton to x:600,y:150,width:200,height:30

SET font to clearFieldsButton to “Poppins”,Style.PLAIN,14

ADD clearFieldsButton to buttonPanel

WHEN clearFieldsButton is clicked:

CALL clearFields() method

CREATE a new JButton named discountButton with text “Calculate Discount”

SET bounds to discountButton to x:30,y:210,width:200,height:30

SET font to discountButton to “Poppins”,Style.PLAIN,14

ADD discountButton to buttonPanel

WHEN discountButton is clicked:

CALL calculateDiscount() method

CREATE a new JButton named upgradeButton with text “Upgrade Plan”

SET bounds to upgradeButton to x:300,y:210,width:200,height:30

SET font to upgradeButton to “Poppins”,Style.PLAIN,14

ADD upgradeButton to buttonPanel

WHEN upgradeButton is clicked:

CALL addUpgradePlan() method

CREATE a new JButton named payDueAmount with text “Pay Due Amount”

SET bounds to payDueAmount to x:600,y:210,width:200,height:30

SET font to payDueAmount to “Poppins”,Style.PLAIN,14

ADD payDueAmount to buttonPanel

WHEN payDueAmount is clicked:

CALL addPayDueAmount() method

CREATE a new JButton named savetoFileButtpn with text “Save to file”

SET bounds to savetoFileButton to x:30,y:270,width:200,height:30

SET font of savetoFileButton to “Poppins”,Style.PLAIN,14

ADD savetoFileButton to buttonPanel

WHEN savetoFileButton is clicked:

CALL savetofile() method

CREATE a new JButton named readFromFilebtn with text “Read from file”

SET bounds to readFromFileBtn to x:300,y:270,width:200,height:30

SET font of readFromFileBtn to “Poppins”,Style.PLAIN,14

ADD readFromFileBtn to buttonPanel

WHEN readFromFileBtn is clicked:

CALL readFromFile() method

```
SET FRAME visible to true


METHOD getYears():
    CREATE a String array named years with size 50
  FOR I from 0 to 49:
Set years[i] to (2025 - i) converted to String
RETURN to years array



METHOD getMonths():
    CREATE a String array named years with size 12
  FOR I from 1 to 12:
Set months[i] to (i+1) converted to String
RETURN to months array



METHOD getDays():
    CREATE a String array named years with size 7
  FOR I from 1to 7:
Set years[i] to (i+1)converted to String
RETURN to months array


METHOD addRegularMember()
```

TRY:

Get ID from idField and convert to integer

IF id already exist:

SHOW error message

GET name, location, phone, email, and referral form text field

IF maleButton is selected:

gender = “male”

ElSE IF femaleButton is selected:

gender=”female”

ELSE IF otherButton is selected:

gender = “other”

ELSE

SHOW ERROR MESSAGE

IF any of the DOB or MembershipStart combo box are not selected

SHOW ERROR MESSAGE

EXIT the method

CONSTRUCT DOB String as “year-month-day” from DOB ComboBox

CONSTRUCT membershipStartDate String as “year-month-day” from membership

Combo box

IF any required text field are empty:

SHOW error message

IF phone is NOT exactly 10 digits:

SHOW Error Message

IF email is NOT in invalid format

SHOW Error Message

CREATE and add the Regular Member

CREATE RegularMember Object with provided details

SHOW success message

CATCH NumberFormatException:

SHOW error message

METHOD addPremiumMember()

TRY:

Get ID from idField and convert to integer

IF id already exist:

SHOW error message

GET name, location, phone, email, and referral form text field

7 IF maleButton is selected:

gender = “male”

ElSE IF femaleButton is selected:

gender=”female”

ELSE IF otherButton is selected:

gender = “other”

ELSE

SHOW ERROR MESSAGE

IF any of the DOB or MembershipStart combo box are not selected

SHOW ERROR MESSAGE

EXIT the method

CONSTRUCT DOB String as “year-month-day” from DOB ComboBox

CONSTRUCT membershipStartDate String as “year-month-day” from membership

Combo box

IF any required text field are empty:

SHOW error message

IF phone is NOT exactly 10 digits:

SHOW Error Message

IF email is NOT in invalid format

SHOW Error Message

CREATE and add the PremiumMember

CREATE PremiumMember Object with provided details

SHOW success message

CATCH NumberFormatException:

SHOW error message

METHOD addactivateMembership()

TRY: Get ID from idField and convert to integer

CALL isValidID and store the result in a variable name member

IF member == null:

SHOW error message

EXIT the method

Call activateMembership() on the member

SHOW success message

CLEAR all input fields

CATCH NumberFormatException:

SHOW error message

METHOD adddctivateMembership()

TRY: Get ID from idField and convert to integer

CALL isValidID and store the result in a variable name member

IF member == null:

SHOW error message

EXIT the method

Call deactivateMembership() on the member

SHOW success message

CLEAR all input fields

CATCH NumberFormatException:

SHOW error message

METHOD addmarkAttendanceBtn()

TRY:

GET ID from idField and convert to integer

CALL isValidID and store the result in a variable name member

IF member is not active

SHOW Error Message

CALL markattendance() on the member

SHOW success message

CATCH NumberFormatException:

SHOW error message

MEHOD addrevertRevertRegularMemberbtn()

TRY:

GET ID from idField and convert to integer

CALL isValidID and store the result in a variable name member

IF member is not active

SHOW Error Message

IF member is NOT instance of Regular Member

SHOW error message

EXIT the Method

GET text from removalField and store in a reason

IF reason is empty

SHOW error message

CALL revertRegularMember on RegularMember

```
SHOW success message

CATCH NumberFormatException:

  SHOW error message


MEHOD addrevertRevertPremiumMemberbtn()

TRY:

     GET ID from idField and convert to integer

      CALL isValidID and store the result in a variable name member

IF member is not active

SHOW Error Message

IF member is NOT instance of Premium Member

SHOW error message

EXIT the Method


GET text from removalField and store in a reason

IF reason is empty

SHOW error message

CALL revertPremiumMember  on PremiumMember

SHOW success message

CATCH NumberFormatException:

  SHOW error message
```

METHOD Display Members()

CREATE a new JFrame named display with text “Member List”

SET defaultCloseOperation to DISPOSE on exit

SET size to width:600, height:600

SET layout to null

CREATE a new JTextArea named displayArea

SET editable of displayArea to false

CREATE a new JSrollPane named scrollPane

SET scrollPane bounds to x: 10,y:10,width:560:height:540

ADD scrollPane to displayPane

IF member is empty:

SET text to “No members found”

CREATE a stringBuilder named as displayText

For each member in the memberlist

IF member is regularMember

APPEND regularMember

IF member is PremiumMember

APPEND premiumMember

APPEND members details like ID, name, location, Phone, Email, Gender, DOB, Start Date, Attendance, Loyalty Points, Active Status

IF member is instance of regularMember:

APPEND plan and price

IF removalReason is not null and removalReason is not empty

APPEND removal Reason

ELSE IF member is instance of premiumMember:

APPEND PersonalTrainer, PaidAmount and Payement Status

IF remainingAmount= preMember * getPremiumCharge

APPEND remainingAmount

IF premium member get fullPayement

APPEND discountAmount

SET displayFrame visible to true

METHOD savetoFIle()

TRY:□ OPEN a filewriter to wrie to “membersDetails.txt”

WRITE header Line with Membership details

Write colums header like ID, Name,location, phone, email, Membership Start date , Plan, Price, Attendance, loyalty Points, Active Status, Full Payment, Discount amount, Paid Amount, Net Amount Paid

IF member is Regular Member:

Write formatted details like ID, Name, location, phone, email, Membership Start date,plan, price, Attendance, loyalty Points, Active status, full payment, discount amount, paid amount, net amount paid

ELSE IF member is Premium Member

Write formatted details in row in order Like ID, Name, location, phone, email, Membership Start date, plan, price, Attendance, loyalty points, Active status, full payment , discount amount, paid Amount, net amount paid

SHOW success message

CATCH NumberFormatException

SHOW error message

METHOD reafFromFile()

CREATE a JTextArea and set editable to false

SET size to rows 20, coloums 80

ADD text area to scrollPane

TRY:

OPEN the file “Membership Details” using a Scanner

WHILE there are more lines in the file

  READ each line and append it to text area

SHOW the scrollable text area in a message dialog titled “Member details”

CATCH IOEXCEPTION

SHOW error message

(Sovereign Corporate Tower, 2025)

Method description

Gym Member

getID(): This is a integer return type method that returns an integer value of the id of the member.

getName(): This is a String return type method that returns an String value of the full name of the member.

getLocation(): This is String return type method that returns the location of the member.

getPhone(): This is a String return type method that return the phone number of the member.

getGender(): This is a String return type method that return the gender of the member.

getDOB(): This a String return type method that return the DOB of the member.

getMembershipStartDate: This is a String type method that returns the date member started their membership.

getAttendance():This is a integer return type that return the attendance of the member.

getLoyaltyPoints():This is a double retutn type that returns the loyalty points of the member.

getActiveStatuss(): This is a Boolean type method that returns the current active status true or false.

markAttendance(): This is an abstract method that must be implement in the base class of the Gym Member.

activateMembership():This method set the active status of Gym Member to true.

deactivateMembership(): This method set the active status of Gym Member to false.

resetMember():This method set the active status to true and set the attendance and loyalty point to 0

displayMember():This method prints the details of Gym Member in the console.

Regular Member

getAttendanceLimit(): This is a integer type method return maximum number of days the regular member is allowed to attend the gym.

isEligibleForUpgrade(); This is Boolean type method return the member qualifies for upgrade of Regular Member.

getReferralSource(): This is String type method return the reason provided to remove the member from the system.

getPlan(): This is a String type method return the member how they referred to the gym.

getPrice(): This is a double type method that return the price for the regular mrmber.

markAttendance(): This method updates a regular member attendance and loyalty member of the premium member.

getPlanPrice(): This method returns price of the membership plan based on the plan name provided.

upgradePlan(): This method attempts to upgrade a regular member current membership plan to new one.

revertRegularMember():This method is used to reset a regular member status when they are removed.

Display():This method prints all the details of premium member in the console.

Premium Member

getPremiumCharge() : Returns the fixed membership amount which is 50000.

getPersonalTrainer(): Returns the name of the personal trainer for the premium Member.

getIsFullPayment(): Returns a Boolean value whether the member made his full payment or not.

getPaidAmount():Returns the total amount of the premium member has paid.

discountAmount(): Returns the discount amount which is applied for premium member.

markAttendance():This method updates a Premium member attendance and loyalty member of the premium member.

payDueAmount(): This method allows a premium member to make a payment towards their premium charge by adding the specified amount of the total paid.Lastly return a message indicating the success of payment along with remaining balance.

revertPremiumMember():This method reset the premium member by calling the parent resetMember method, remove the personal trainer information, set full payment to false and set discount amount to 0.

Display(): This method call the parent class display method, and print all the details of premium member.

Gym GUI

getYears(): This method creates an array of 50 years starting from 2025 and returns the current year.

getMonths(): This method creates an array of months as String form 1 to 12 and returns the current month.

getDays():This method creates an aray of 30 days as String and returns the current day .

addRegularMember(): This method handles the addition of a new regular gym member. Firstly It check the proper validation of the id and id should be unique and must verify required fields such as name, location, phone number, email and referral source and the field could not be empty. The method ensured the phone number should be exactly with 10 digits and ensures the proper validation of the email. If all the validation is check it create a new regular member object when the user input all the data and adds to the member list.

addPremiumMember(): This method handles the addition of a new regular gym member. Firstly It check the proper validation of the id and id should be unique and must verify required fields such as name, location, phone number, email and referral source and the field could not be empty. The method ensured the phone number should be exactly with 10 digits and ensures the proper validation of the email. If all the validation is check it create a new Premium member object when the user input all the data and adds to the member list.

activateMembership(): This methods actives the membership status of the member. Firstly it parse the ID input and check if the ID is already exists using inValidID method. If the member is found that the membership set to active by calling the activatemembership method and display a success method. If the id is nt found then it give the error message.

adddeactivateMembership():This methods actives the membership status of the member. Firstly it parse the ID input and check if the ID is already exists using inValidID method. If the member is found that the membership set to active by calling the deactiveMembership method and display a success method. If the id is nt found then it give the error message.

addMarkAttendancebtn():This method is used for the mark attendance of the gym member. This method first validated the input ID and check whether the id is valid or not using isValidId method. If the member is found and their membership is active then it call the markAttendance() method to keep record for the attendance and calculate loyalty points.

addRevertRegularmemberbtn(): This method reverts a Regular member to the initial state. This method first validates the input ID and check whether the id is valid or not using isValidID method. It check it is of Regular Member and verifies the removal reason has entered or not. If all condition are meet then it call the revertRegulatMember method with a given reason to reset the member attributes.

addRevertPremiumMember():This method reverts a Regular member to the initial state. This method first validates the input ID and check whether the id is valid or not using isValidID method. It check it is of Premium Member and verifies the removal reason has entered or not. If all condition are meet then it call the revertPremiumMember method with a given reason to reset the member attributes.

displayMembers():This methods create a new frame to display details of gym member. It uses a JTextArea inside a JScrollPane. The method first validates the input of ID and it checks it is valid id or not using isValidID method. If the member list is empty it show a message “No members found”. If the member is instance of premium member and regular member it display the members details like Id, name, contact details, membership status ,plan and payment details.

clearFields():The method reset all the input fields to their default state. It clear all the text from the field like in idField, nameField, locationField, emailField, referralField, paymentField and trainerField. It also reset the DOB and membership Start date of JCombboBox and also clear the selected gender of JRadioButton.

calculateDiscount(): This method handles the discount amount of the premium member. It first handles the input ID and check it is valid id or not. Check the id already exist or not in premium member. If valid then premium charge makes the payment using payDueAmount() method and implement discount in the premium member.

addUpgradePlan(): This method allow the regular member to upgrade the plan. Firstly it check the input id is valid or not using isValid method. It checks whether the member has met the attendance requirement for the upgrade. If this condition match then this method select new plan. Then it attempts the upgradePlan method and display the outcome through the dialog box.

addPayDueAmount(): This method allows a premium member to pay the remaining membership charge. Firstly it check the input id and check is this is valid or not. If valid then it receive the premium charge amount from the input field and process the payment using payDueAmount method. If all the process run successfully a confirmation dialog is shown.

saveToFIle(): This method take all the information from gym members and store the data in “Members.txt” file. Firstly it write the header with in an formatted columns. And lastly write all the details of premium member and regular member in a well structured row. For regular member id, name, contact, membership start date, plan, price, attendance, loyalty points and active status is saved in the txt file. And for premium member all the regular member details and and additionally payment status, discount amount, paid amount and net amount are included.

readFromFile(): This method reads and display members of gym member stored in the Members.txt file. It ensures that saved member data can be viewed directly from the file in a readable format.

2 Testing

Test 1

Compile and run the program using command prompt

PROCESS

OUTPUT

Objective

To Compile and Run program using from Cmd.

Action

Opened Command Prompt, Compiled the code using JavaC prompt then the code was executed.

Excepted Output

The GymGUI program should Run.

Actual Output

GymGUI program run successfully.

Result

The test was successful.

Table 6:Test 1

Figure 4: Test 1

Figure 5: Test 1 Output

Test 2

Add Regular Member and Premium Member

PROCESS

OUTPUT

Objective

Adding Regular member and premium member respectively.

Action

Regular Member was added first and then premium member was added.

Excepted Output

Both type of members should be added successfully.

Actual Output

Both type of members was added successfully.

Result

The Test was Successful.

Table 7:Test 2

Figure 6:Test 2 Output

Figure 7:Test 2 Output

Test 3

Test case for mark Attendance

PROCESS

OUTPUT

Objective

Test Case for Mark Attendance and upgrade plan

Action

Regular Member was added first then membership was activated and then attendance was marked and then displayed the member details.

Excepted Output

The added member attendance should be marked and should be displayed 1.

Actual Output

The added member attendance was marked and displayed as 1

Result

The Test was Successful.

Table 8: Test 3

Figure 8: Test 3 Output

Figure 9:Test 3 activate membership

Figure 10:Test 3 mark Attendance

Test case for Upgrade plan

PROCESS

OUTPUT

Objective

To upgrade the plan of regular member

Action

Fill all the required fields and add regular member and set membership active and the attendance should be more than 30

Excepted Output

The plan may be selected to standard.

Actual Output

The plan selected to standard.

Result

14 The test was successful.

Table 9: Test 3 upgrade plan

Figure 11: test 3 adding regular member for upgrade plan

Figure 12:Test 3 active membership for upgrade plan

Figure 13:Test 3 marking attendance 38 for upgrade plan

Figure 14:test 3 successfully upgrade the plan

Test 4

Test case for calculate discount

PROCESS

OUTPUT

Objective

To calculate the discount for premium member.

Action

Fill all the required fill and add to premium member and paid total amount to get discount

Excepted output

May give 10 percent discount

Actual Output

Give 10 percent discount

Result

The test was successful.

Figure 15: Test 4 adding premium member for discount

Figure 16:Test 4 output of discount amount

Test case for pay due amount

PROCESS

OUTPUT

Objective

To test the premium member can successfully pay the due premium amount.

Action

Fill all the required fields, registers a premium member and click the pay due amount.

Excepted output

May paid due amount

Actual Output

The paid due amount has been paid

Result

The test was successful.

Table 10: test case for pay due amount

Figure 17:test case of adding premium member for pay due amount

Figure 18:Test case for pay due amount output

Test case for revert regular member

PROCESS

OUTPUT

Objective

2 To verify to revert regular member

Action

Fill all the required fields, and add a regular member, then enter the revert regular member button.

Excepted Output

The member should be successfully reverted.

Actual Output

The member successfully reverted.

Result

The test was successful.

Table 11: Test case for revert regular member

Figure 19: Adding regular member for regular member test case

Figure 20:Test case for revert regular member output

Test case for Premium Member

PROCESS

OUTPUT

Objective

To verify revert Premium Member

Action

Fill all the required field and add a premium member then enter the revert premium member button.

Excepted Output

The member should be successful reverted.

Actual Output

The member successful reverted.

Result

The test was successful.

Table 12: Test case for premium member

Figure 21: Adding premium member for test case for premium member

Figure 22:Test case for revert regular member

Test case for save to file

PROCESS

OUTPUT

Objective

To verify the system correctly saved the details of member

Action

Add both premium member and regular member and click the save to file button

Excepted Output

A file name should be created and all the data should be stored there.

Actual Output

A file created and stored all the data there

Action

The test was successfull

Table 13: Test case for save to file

Figure 23:Adding regular member for test case of save to file button

Figure 24:Adding premium member for test case for save to file

Figure 25:Click save to file button

Read from file

PROCESS

OUTPUT

Objective

To read the file from the text file.

Action

Add both regular member and premium member , click save to file button and lastly click read from file button

Excepted Output

May stored the data can be seen.

Actual Output

Stored data can be seen.

Result

The test was successful.

Table 14:Test case of Read From file

Figure 26:Adding regular member for read from file test

Figure 27:Adding premium member for test case for read from file

Figure 28:Output of test case of read from file

Error detection and correction

Syntax error

Syntax error in java means mistake of programmer while writing the code occurs when compiler cannot understand the source code

Figure 29:Detecting syntax error

(Burnham, 2025)

Figure 30:Solving syntax error

Run time error

Run time error means that error when a program doesn't contain any syntax error but it occurs during the execution of the program.

Figure 31:Detection of run time error in code

Figure 32;Detection of run time error in GUI

Figure 33:Correction of run time error in code

Figure 34:Correction of run time error in GUI

(Burnham, 2025)

Logical error

Logical error is a mistake in a source code that result unexpected behavior. It is a type of run-time error it means it occurs during the execution of the program.

Figure 35:Detection of logical error code

Figure 36:Correcting the logical error

Conclusion

In this project, I successfully developed a Gym Management System using Java and using object oriented principal concept. The system includes one parent class named Gym Members and sub class premium member and regular member. I have also created a GUI using java swing.

While working on this assignment I have learned how to use Java concept like abstraction, inheritance, polymorphism. I have also work with Array List and also create a user friendly GUI using Java Swing and also implement all the exception in the program.

During the development of the system I have faced several issues like handling user input fields, handling errors, and mainly I have struggled to organize the components using layout manager to null.

To solve the problem I start debugging the program, used try-catch block for error handling and for organize components I took a notes copy and first run dry program in my notes copy which helps me to organize component easily.

Overall the project was a great learning experience that build by Java Learning experience that build my java knowledge in a real word scenario.

References

5 Burnham, D., 2025. /tutorial/david/errors-in-java. [Online] Available at: https://codehs.com/tutorial/david/errors-in-java [Accessed 15 05 2025].

5 Burnham, D., 2025. tutorial/david/errors-in-java. [Online] Available at: https://codehs.com/tutorial/david/errors-in-java [Accessed 15 05 2025].

10 Sovereign Corporate Tower, 2025. how-to-write-a-pseudo-code/. [Online] Available at: https://www.geeksforgeeks.org/how-to-write-a-pseudo-code/ [Accessed 15 05 2025].

13 Sovereign Corporate Tower, 2025. unified-modeling-language-uml-class-diagrams/.

[Online]

Available at: https://www.geeksforgeeks.org/unified-modeling-language-uml-class-diagrams/

[Accessed 15 05 2025].